



```
def store_target(state):
```

```
...
```

```
state = execute(flip(state, angle=pi / 2))
```

```
displacement = storage_displacement(state)
```

```
state = execute(press_move(state, displacement=displacement))
```

```
return state
```

 Π

```
def flip(state, angle):
```

```
def cost(traj):
```

```
return (
```

```
angle_error(traj[-1], angle)
```

```
+ pivot_drift(traj)
```

```
+ contact_error(traj[-1], state.fixture)
```

```
+ support_violation(traj, state.support)
```

```
+ motion_roughness(traj)
```

```
)
```

```
return local_traj_opt(state, cost)
```

 P_{k-1}

```
def press_move(state, displacement):
```

```
goal = state.target.position + displacement
```

```
def cost(traj):
```

```
return (
```

```
position_error(traj[-1], goal)
```

```
+ orientation_error(traj, state.target)
```

```
+ tool_contact_loss(traj.moving)
```

```
+ support_violation(traj, state.support)
```

```
+ motion_roughness(traj)
```

```
)
```

```
return local_traj_opt(state, cost)
```

 P_k
 s_{k-1}

```
def storage_displacement(state):
```

```
axis = along(state.fixture)
```

```
offset = state.storage_reference.position - state.target.position
```

```
return dot(offset, axis) * axis
```

 c_k
 g_k